

# SML Assignment 1

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## 1. On the Nine Wheels of SML

### **Problem Chosen: Predicting House Prices for a district in Taiwan.**

The real estate market plays a key role in any economy, as it reflects the combined effects of demographics, the economy, and location. Knowing the likely price of a house is important, not just for buyers and sellers trying to make fair deals, but also for policymakers and city planners who want to support sustainable and accessible housing. Many factors can affect house prices. For example, the age and condition of a property, how close it is to public transport or shops, the local economy, and the neighborhood itself all make a difference. Houses near transit stations or popular commercial areas usually sell for more, while older houses or those in less convenient locations tend to be cheaper. In Taiwan, fast urban growth and high population density make reliable property valuations even more important. By looking at things like demographics, location, and economic indicators, it is possible to estimate house prices more accurately. These estimates can help everyone, from homebuyers to government officials, make better decisions about real estate.

Building on this, our project aims to develop a system that predicts house prices per unit area for properties in a district in Taiwan. The system will use key features such as the age of the house, proximity to the nearest MRT station, the number of nearby convenience stores, and the geographic coordinates of the property. These features are particularly important because they capture the factors that most strongly influence a property's value: house age reflects its condition and potential maintenance costs, proximity to public transport affects accessibility and convenience, nearby amenities indicate how desired the area is, and geographic location captures broader neighborhood effects that might not be visible from other data alone. By focusing on these features, we can build a model that provides accurate, continuous-valued price estimates to support informed decisions in the housing market.

### **Wheel 1: Clear Problem Choice and Formulation**

#### **Specific Choice/Action:**

We formulate the task as a regression problem, where the goal is to predict

$$Y = \text{house price per unit area} \in [7.6, 117.5]$$

.

given a feature vector

$x$  = (transaction date, house age, distance to MRT station, number of convenience stores, latitude, longitude).

#### **Thorough Justification:**

The target variable (Price) is continuous, making regression the natural choice. Regression captures housing price details, allowing more accurate estimates for everyone. Predictions are expected to lie roughly within the range of observed prices in the dataset, though the model could, in principle, produce values outside this range. This makes a regression formulation the most natural thing to choose.

## Wheel 2: Data Collection and Exploration

### Specific Choice/Action:

We use the public dataset from the UCI Machine Learning Repository.

It contains 414 observations and 6 attributes, including transaction date, house age, distance to the nearest MRT station, number of convenience stores, latitude, longitude, and house price per unit area. These features form our input space,  $\mathcal{X} = \mathbb{R}^6$ , with dimension 6. The index  $\kappa = 69$  indicates our dataset is an overdetermined system thus indicating stability of the regression estimates. The output space,  $\mathcal{Y}$  is a subset of the real line  $\mathbb{R}$

We will perform the following exploratory data analysis (EDA) steps:

1. Inspect the distribution and range of each feature.
2. Check for missing values and handle them if any.
3. Examine pairwise correlations to identify multicollinearity.
4. Visualize relationships between features and the target variable using scatterplots and boxplots.

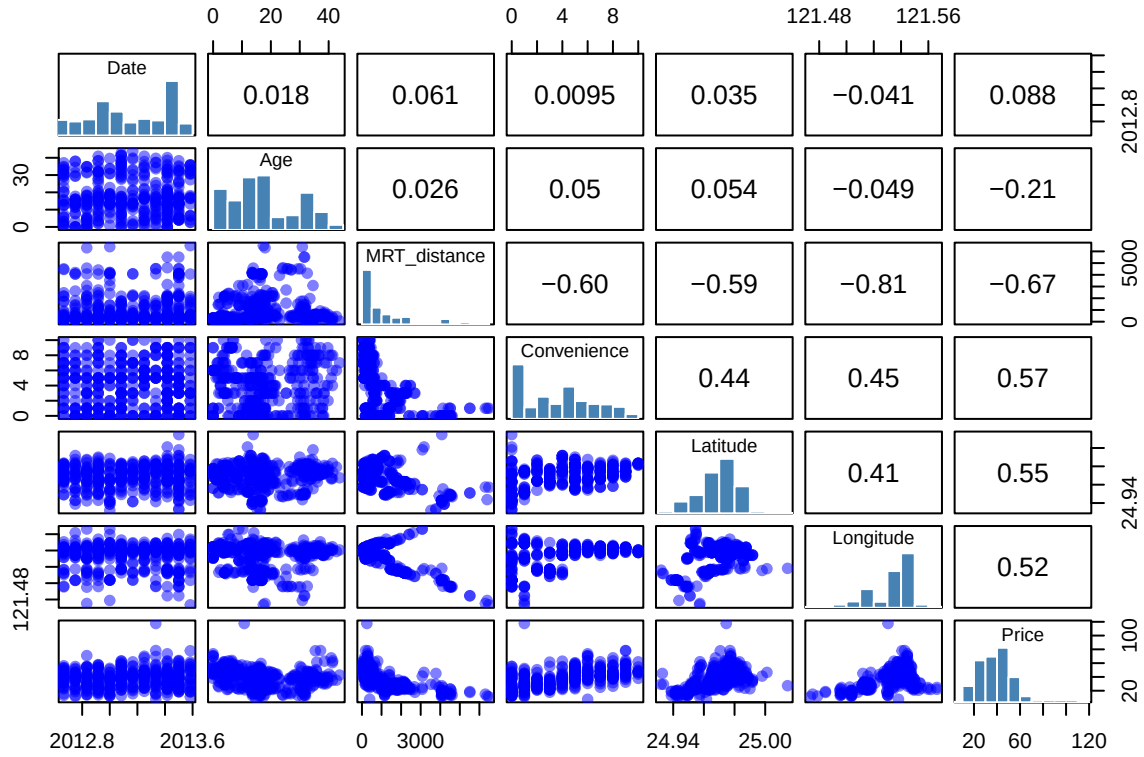
**Thorough Justification:** The UCI repository is useful in the situation since it contains historical data on house prices, readily available with all the necessary features. Exploring the data is essential for understanding patterns, identifying outliers, and guiding model selection. Correlation analysis helps detect multicollinearity, which can affect model stability in linear regression. Proper EDA ensures that our model is grounded in the structure of the data and that any preprocessing or feature engineering decisions are well-informed.

Table 1: Summary statistics for Real Estate dataset

|              | vars | n   | mean        | sd           | median    | min        | max        |
|--------------|------|-----|-------------|--------------|-----------|------------|------------|
| Date         | 1    | 414 | 2013.148953 | 0.2819953    | 2013.1667 | 2012.66667 | 2013.58333 |
| Age          | 2    | 414 | 17.712560   | 11.3924845   | 16.1000   | 0.00000    | 43.80000   |
| MRT_distance | 3    | 414 | 1083.885689 | 1262.1095954 | 492.2313  | 23.38284   | 6488.02100 |
| Convenience  | 4    | 414 | 4.094203    | 2.9455618    | 4.0000    | 0.00000    | 10.00000   |
| Latitude     | 5    | 414 | 24.969030   | 0.0124102    | 24.9711   | 24.93207   | 25.01459   |
| Longitude    | 6    | 414 | 121.533361  | 0.0153472    | 121.5386  | 121.47353  | 121.56627  |
| Price        | 7    | 414 | 37.980193   | 13.6064877   | 38.4500   | 7.60000    | 117.50000  |

**Brief interpretation:** The dataset contains 414 observations per variable (no missing values) with six input features and the target variable, **house price per unit area**. The features are type homogeneous but scale heterogeneous. For example, the distance to the nearest MRT station ranges from about 23 to over 6,400 meters, while the number of nearby convenience stores goes from 0 to 10. Even though latitude and longitude change only slightly, they can still affect property prices. Comparing the mean and median values shows some skewness in certain features, especially MRT distance and house price, suggesting the presence of outliers or extreme values.

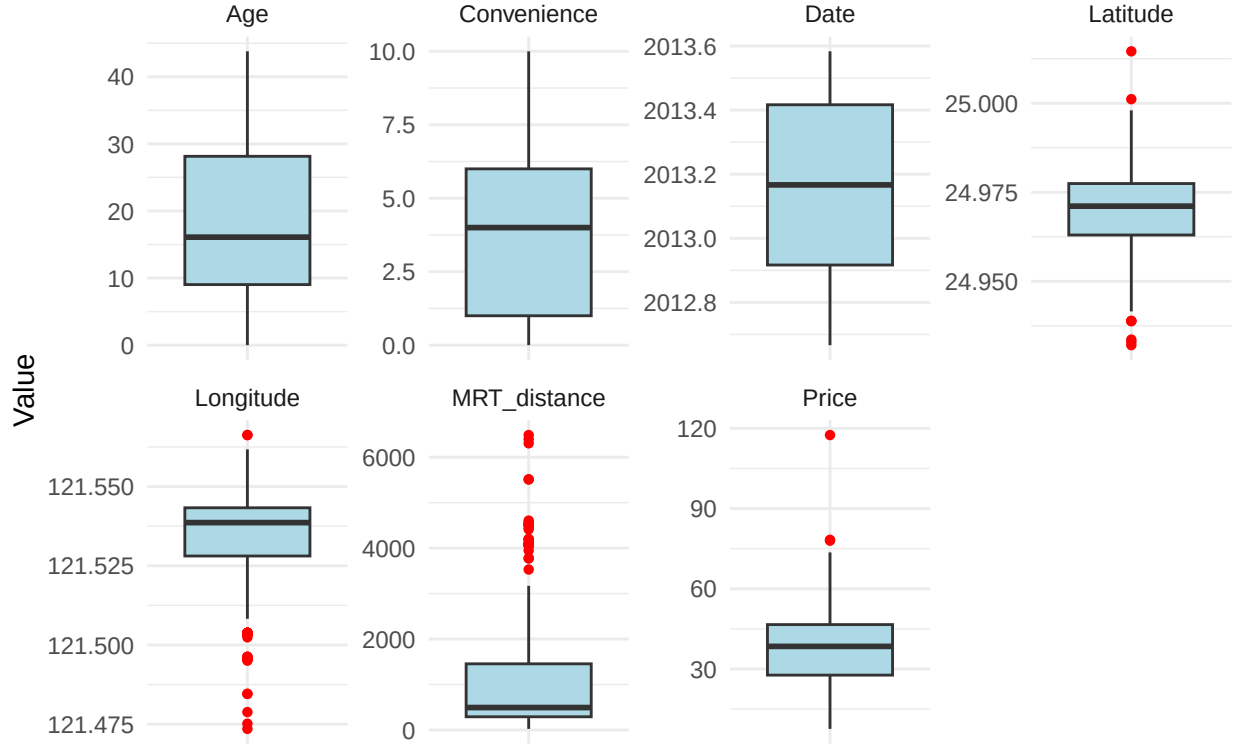
For the regression task, this means a few things. The features may need to be scaled so that differences in units don't skew the model. Handling outliers or using robust regression methods could improve stability. Distance and location features might also benefit from transformations, such as taking the logarithm, to better capture their impact on house prices.



**Brief interpretation:** The correlation analysis shows that some features have stronger relationships with house **Price** than others. In particular, **MRT\_distance** is negatively correlated with price (around -0.67), meaning houses closer to MRT stations tend to be more expensive. **Convenience**, **Latitude**, and **Longitude** show moderate positive correlations with price, suggesting that location and nearby amenities contribute to higher values. Other features, like **Date** and **Age**, show weaker correlations, indicating a smaller direct effect on **Price**.

The scatterplots reveal some non-linear patterns, especially for **MRT\_distance** and **Convenience**, and potential outliers are visible in several variables. This suggests that some feature engineering could improve model performance. For instance, log-transforming distances, creating interaction terms between location features, or encoding spatial information more effectively. Scaling features will also help since the variables are on very different ranges.

## Boxplots of Features and Target Variable



### Brief Interpretation:

The boxplots support what we already noticed from the correlation analysis. Location and accessibility features don't vary much, but even those small differences still reflect meaningful neighbourhood patterns. In contrast, features like Age and MRT distance spread out more and show clear outliers. Price is also right-skewed, with a few properties selling for much more than the rest. Overall, the distributions suggest that scaling the features and being mindful of outliers will help the model, but the dataset remains clean and very suitable for regression.

### Wheel 3: Hypothesis Space

In statistical learning, the hypothesis space  $\mathcal{H}$  specifies the set of functions the learning machine is allowed to search over. This choice determines the bias-variance tradeoff: simpler spaces give stable models which are more likely to underfit, while richer spaces offer flexibility at the cost of higher variance. Guided by this principle and based on the information from the scatter plots, I restrict my hypothesis space to three classes of regression functions for predicting house prices.

#### 1. Linear Regression Class $\mathcal{H}_{\text{lin}}$

$$\mathcal{H}_{\text{lin}} = \{ f(X) = X^\top \beta \}.$$

This space corresponds to classical linear regression. It is high-bias but low-variance. Its coefficients give information about how each feature affects **Price**.

#### 2. Polynomially Enriched Class $\mathcal{H}_{\text{poly}}$

$$\mathcal{H}_{\text{poly}} = \mathcal{H}_{\text{lin}} \cup \{ \gamma_1 \text{Age}^2 + \gamma_2 \text{MRT\_distance}^2 \}.$$

This choice of functions is based on the patterns observed in the scatter plots. Some curvature is obvious in the plots of **Price** against each of **Age** and **MRT\_distance** even though some parts of the relationship with **Age** looks linear.

This structured progression aligns with the Wheel 3 philosophy: choose a model space that is “as simple as possible, but not simpler.” It enables systematic comparison in later wheels and maintains interpretability, which is essential in real estate valuation.

#### **Wheel 4: Principle of Learning and Criteria**

Now that we have picked our  $\mathcal{H}$  in Wheel 3, the next thing to look at is how to pick the “best” function out of that space? In statistical learning, the answer for a regression task is Empirical Risk Minimization, or ERM for short. Since we do not know the true distribution from which our dataset is sampled, we cannot directly compute the true risk, given by

$$R(f) = \mathbb{E}_{(X,Y) \sim p} [(Y - f(X))^2],$$

of any model. What the Learning Principle does is to use the data and then calculate the average error over all these values in the dataset. Our best learning machine is the one that makes this empirical risk as small as possible. Mathematically this empirical risk is given by

$$R_n(f) = \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2.$$

In context the Empirical Risk Minimization is just taking the learning machine whose predictions are as close as is allowed (without overfitting) to the true values. The squared error loss is used here due to the fact that for this loss function larger mistakes are penalized more heavily than smaller ones, ensuring that the model focuses on avoiding big mispredictions. Additionally since the squared error loss is differentiable everywhere then the computations related to finding the best learning machine using this loss function become easier to handle.

#### **Wheel 5: Algorithmic and Computational Optimization**

#### **Wheel 6: Regularization and Refinement**

#### **Wheel 7: Extrinsic Predictive Comparisons**

#### **Wheel 8: Statistical Inference and Theoretical Justification**

#### **Wheel 9: Deployment and Practical Scalability**

Deployment is where the model is made available for use in the sense that it is used for its intended purpose. In the House Price context this means that this model is made available for use by stakeholders and all parties involved. This is where the model’s true generalizability is tested. For this project, the code and workflows for this project can be accessed [here](#).

## **2. Practical Machine Learning - Digit Recognition**

### **1. Data Preparation**

In this section the focus was to perform binary classification on a subset of the MNIST digit dataset. The dataset was prepared and a subset of the data, those whose labels are 1(positive class) and 7(negative class), was used for this classification task. Since the MNIST data already comes in a test set and a train set, those were the partitions that were used. Due to the large size of the training data, only 10% of the training data was used to train the model. In this task we classify the digits using the k-Nearest Neighbors (kNN) algorithm. The MNIST training and test labels were filtered for only the numbers 1 and 7. The main training set then was of size 13007 while the stratified sample contained 1300 samples. The number of features in

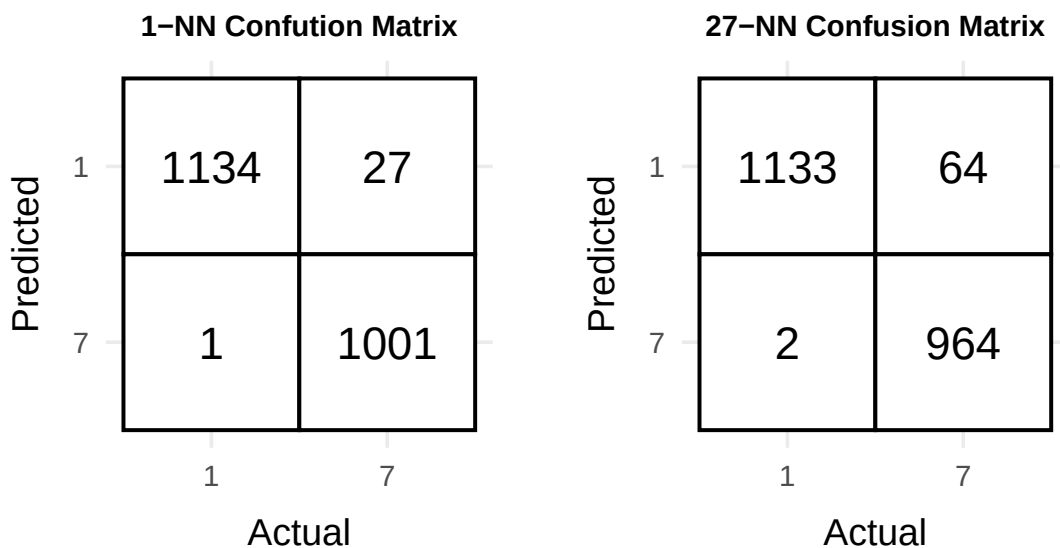
the data set is 784. The choice of using a stratified sample for training was to make training faster, while maintaining class distribution occurring in the original training data.

## 2. Model Training

In this section we train two  $k$ -Nearest Neighbors classifiers based on the stratified sample of the training data. The choice of the kNN classifiers was due to that fact that these models are simple and can handle high dimensional data effectively.

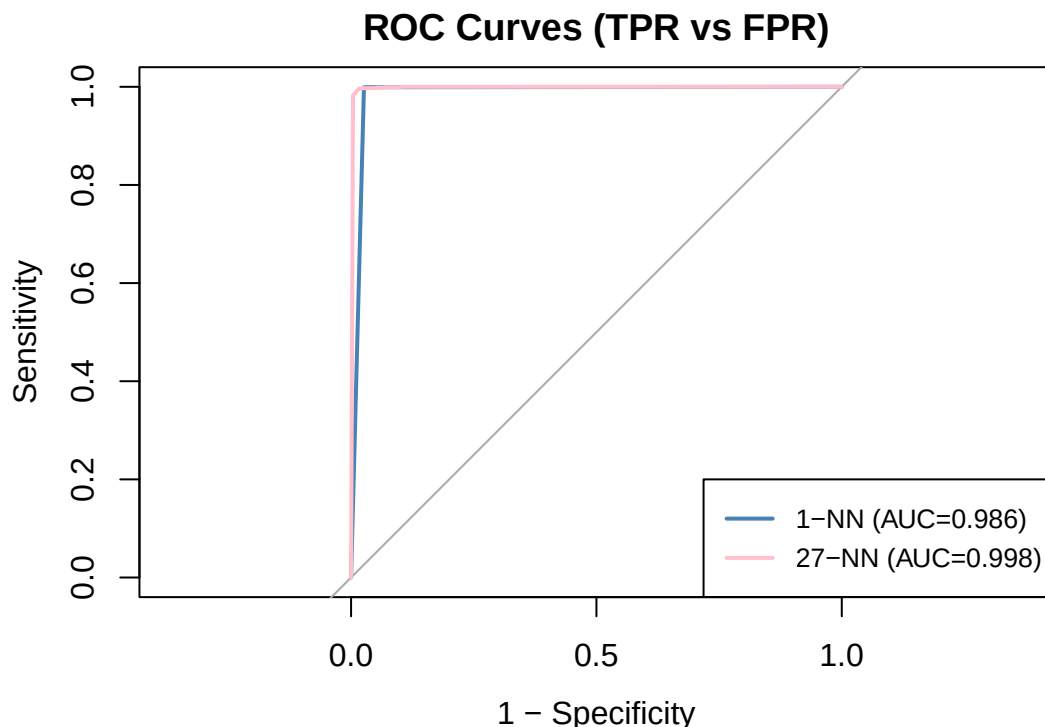
## 3. Performance Metrics

The accuracy and confusion matrix for each of these models is given below.



Based on the confusion matrices, the 1-NN classifier seems to be performing better with 2135 images being correctly predicted while 28 are being wrongly predicted as opposed to the 27-NN where 2097 images being correctly predicted while 66 are being wrongly predicted. The accuracy for the 1-NN model is 98.7055 while the accuracy for the 27-NN model is 96.9487 which is consistent with what is being displayed in the scatter plots.

#### 4. ROC Curve Analysis



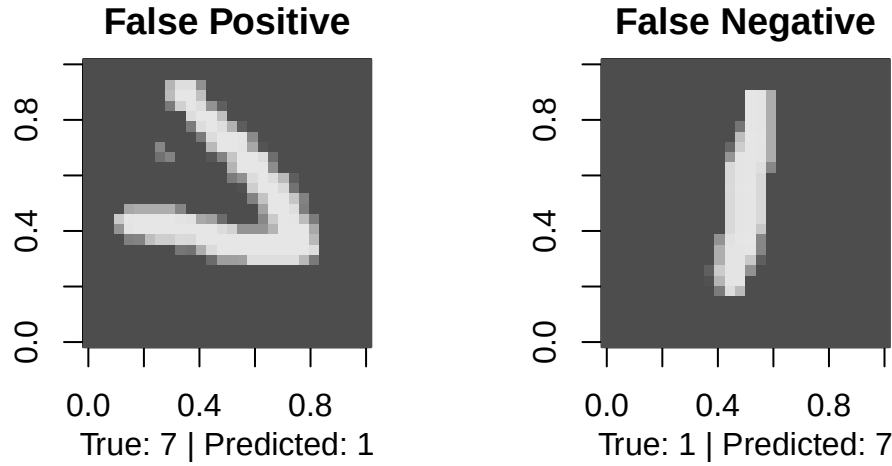
The ROC curve plots the **True Positive Rate**(Sensitivity) against the **False Positive Rate** (1-Specificity) for the two  $k$ -NN models that were trained. In particular both models perform well at correctly classifying images as 1 since the ROC curves for both models lie well above the diagonal, indicating high sensitivity. The 27-NN model shows a slightly higher sensitivity at the same false positive rate as compared to the 1-NN model, suggesting that the former is better at correctly identifying class 1 images. This is consistent with the **Area Under the Curve**(AUC) values indicated in the legend.

While the confusion matrix shows that the 1-NN model correctly predicts more instances in each class than the 27-NN model, resulting in higher accuracy at the chosen threshold, the ROC curves tell a slightly different story. The AUC for the 27-NN model is higher than that of 1-NN, indicating that, across all possible thresholds, the 27-NN model has a slightly better ability to discriminate between the classes. This highlights an important distinction: accuracy from the confusion matrix evaluates performance at a single threshold, whereas AUC summarizes the model's discriminative ability across thresholds.

#### 5. Error Visualization

In this section the plot of some false positives and false negatives, as observed from the model predictions are displayed.

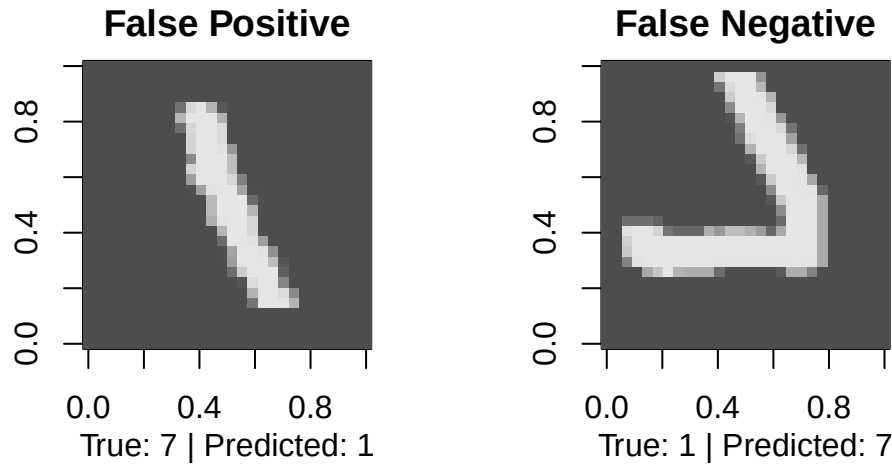
### 1-NN Model Misclassifications



For the 1-NN model we see some examples of false positives and false negatives. In particular we see that the model is predicting the image on the left as a 1 whereas the true value is obviously 7. The model may have made this mistake particularly because of the orientation of the image. In a likewise manner for the false negative where the true label is 1 while the model is predicting a 7. This could be due to the fact that the image looks cropped at the top.

For the example of the false positive for 27-NN model, the reason for this prediction could be in the fact the image looks cropped at the bottom, whereas for the false negative, the prediction could be due to the orientation as well as the cropped nature towards the top of the image. The images are given below.

### 27-NN Model Misclassifications



## 3. Discovering the concept of ultra high dimensionality

For this section 6 different datasets were used. For each dataset, a brief overview of its dimensionality, sample size, and homogeneity is provided, considering both type-homogeneity (consistency of variable types) and scale homogeneity (consistency of measurement scales). The  $n/p$  ratio (number of observations over the number of features) is also computed. Finally, graphical summaries of a subset of variables are provided,

and correlations and multicollinearity are examined where applicable, with the aim of understanding the structure and properties of each dataset.

The general outline for this section is as follows:

1. Download the dataset and open it. In this section the dataset is loaded and displayed to provide some understanding of its contents and structure.
2. Provide a succinct description of the dimensionality of the data, sample size and homogeneity. In this section some comments are made on the dimensionality of the data, its sample size the weather the features are homogeneous and if the scales are homogeneous as well.
3. Commenting on the quintessentially important index  $\kappa := n/p$ . This section gives an insight into the nature of the dimensionality of the data; whether the data is high dimensional or not.
4. Generating basic graphical summaries Some basic graphical summaries like the distribution of the features are given for only 9 of the variables in each dataset.
5. Discussions on correlations and multicollinearity Some comments are made on multicollinearity and correlations between these 9 variables.

## DNA Data

1. Below is a table of a few of the rows in the dataset.

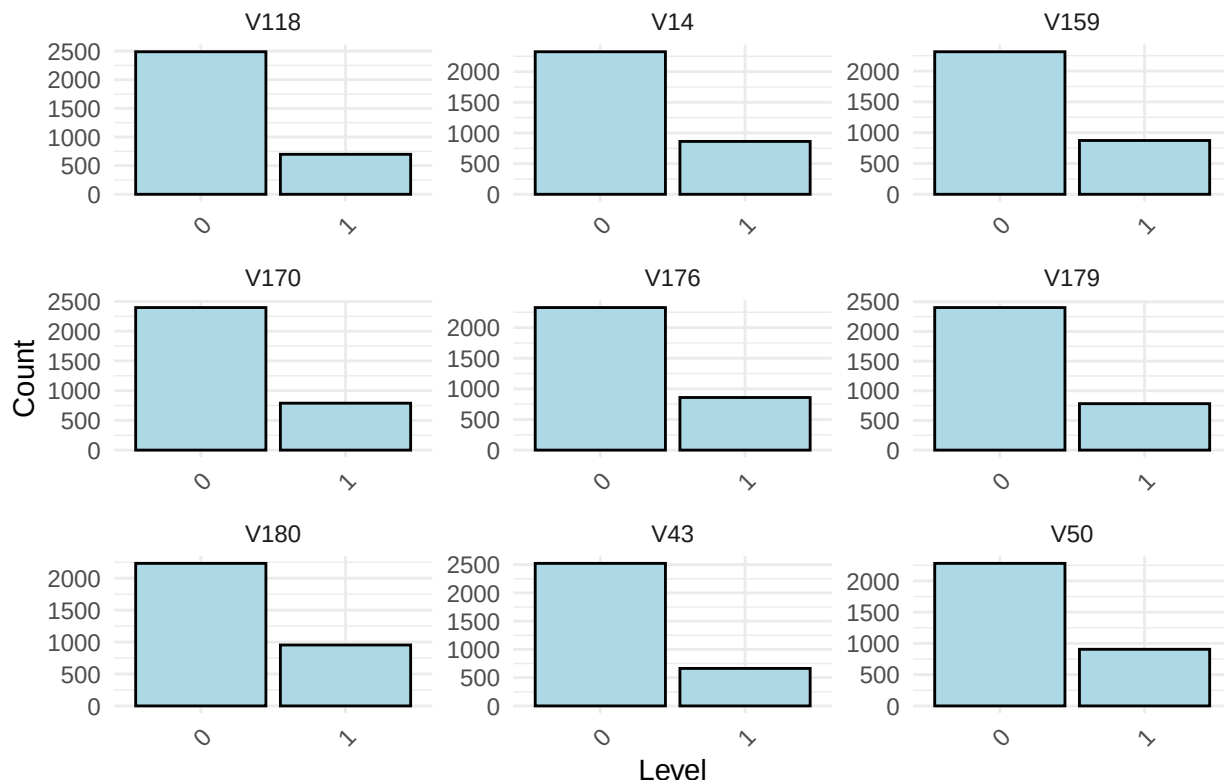
| V159 | V179 | V14 | V170 | V50 | V118 | V43 | V180 | V176 |
|------|------|-----|------|-----|------|-----|------|------|
| 0    | 0    | 0   | 1    | 1   | 0    | 0   | 0    | 0    |
| 0    | 1    | 0   | 1    | 0   | 0    | 0   | 0    | 0    |
| 0    | 0    | 0   | 1    | 0   | 0    | 0   | 1    | 0    |

**2.** This dataset has 3186 observations and 180 features. The features are of 1 different data types (Categorical) among the features. Even though the features look like integer values, they are essentially categorical values that have been encoded, possibly with the usual `label encoding`. In that regard it does not make sense to talk about scale homogeneity.

**3.** The index  $\kappa$  for this dataset is 17.7 this indicates that the number of samples are more than the observed features (an overdetermined system). This suggests that machine learning estimators are likely to be stable, and the risk of overfitting is relatively low. The large  $\kappa$  value implies there is sufficient information in the data to reliably estimate model parameters.

**4.** In this case a high class imbalance is observed for all the 9 sampled categories, indicating that some feature categories dominate the others. In such a situation this can reduce the model's ability to learn from minority categories and limit the contribution of those features to predictive performance.

## Bar Plots of 9 Factor Variables



5. All predictors here are categorical, so Pearson correlation and traditional multicollinearity do not apply. Although categorical variables may still be associated (e.g., via Cramér's V), such associations do not introduce the  $X^T X$ -related instability seen in numerical regression models. Since we are performing multi-class classification, this is not a practical concern for our analysis.

## Breast Cancer Data

1. Below is a table of a few of the rows in the dataset.

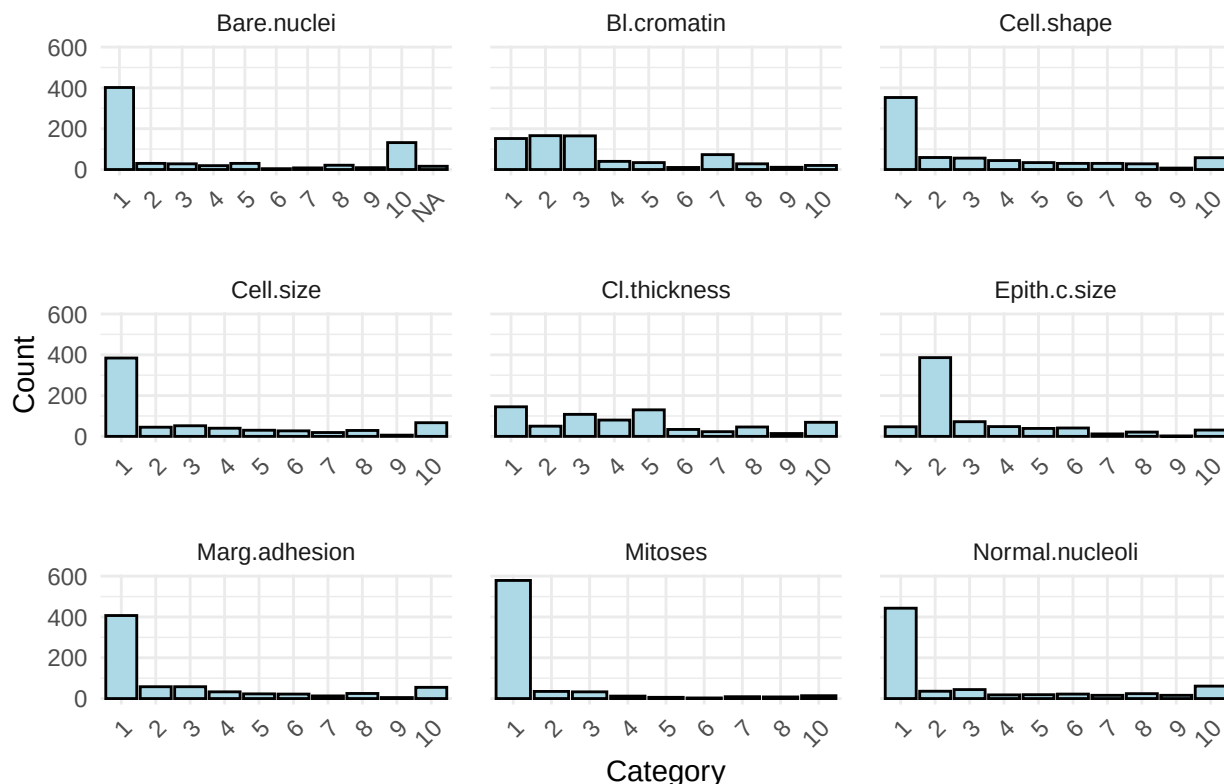
| Cell.size | Class  | Cl.thickness | Epith.c.size | Mitoses | Marg.adhesion | Cell.shape | Normal.nucleoli | Bl.cromatin |
|-----------|--------|--------------|--------------|---------|---------------|------------|-----------------|-------------|
| 1         | benign | 5            | 2            | 1       | 1             | 1          | 1               | 3           |
| 4         | benign | 5            | 7            | 1       | 5             | 4          | 2               | 3           |
| 1         | benign | 3            | 2            | 1       | 1             | 1          | 1               | 3           |

2. This dataset has 699 observations and 9 features. The features together with the response are all categorical (encoded to numeric). The variables are, therefore, type homogenous. Since the features are categorical, it does not make sense to talk about scale homogeneity as the encoded values are really just discrete values. So the notion of scale homogeneity does not apply in the previous sense, although different variables have different numbers of classes.

3. The index  $\kappa$  for this dataset is 61.545 this indicates that the number of samples are more than the observed features (an overdetermined system). This suggests that machine learning estimators are likely to be stable, and the risk of overfitting is relatively low. The large  $\kappa$  value implies there is sufficient information in the data to reliably estimate model parameters.

4. These bar plots show that almost all nine predictors are heavily concentrated in the lowest category, with very few observations falling into higher abnormality levels. This strong skew suggests that most patients present with mild cellular features, while severe morphological changes are relatively rare in the dataset. Such imbalanced category distributions may influence how models learn patterns, since the majority of information comes from the lower-grade classes.

### Bar Plots of 9 Categorical Variables



5. Although the response is binary, all predictors are categorical. This means multicollinearity diagnostics like Pearson's correlation are not appropriate because the model uses dummy-coded factors rather than continuous numeric predictors. Categorical predictors may still show associations (e.g., via chi-square or Cramér's V), but these do not produce the linear-dependence (singularity of  $X^T X$  matrix) issues seen in numeric regression models. Therefore, multicollinearity is not a practical concern here.

### Spam Data

1. Below is a table of a few of the rows in the dataset.

| num650 | project | internet | re   | capitalLong | business | order | free | font |
|--------|---------|----------|------|-------------|----------|-------|------|------|
| 0      | 0       | 0.00     | 0.00 | 61          | 0.00     | 0.00  | 0.32 | 0    |
| 0      | 0       | 0.07     | 0.00 | 101         | 0.07     | 0.00  | 0.14 | 0    |
| 0      | 0       | 0.12     | 0.06 | 485         | 0.06     | 0.64  | 0.06 | 0    |

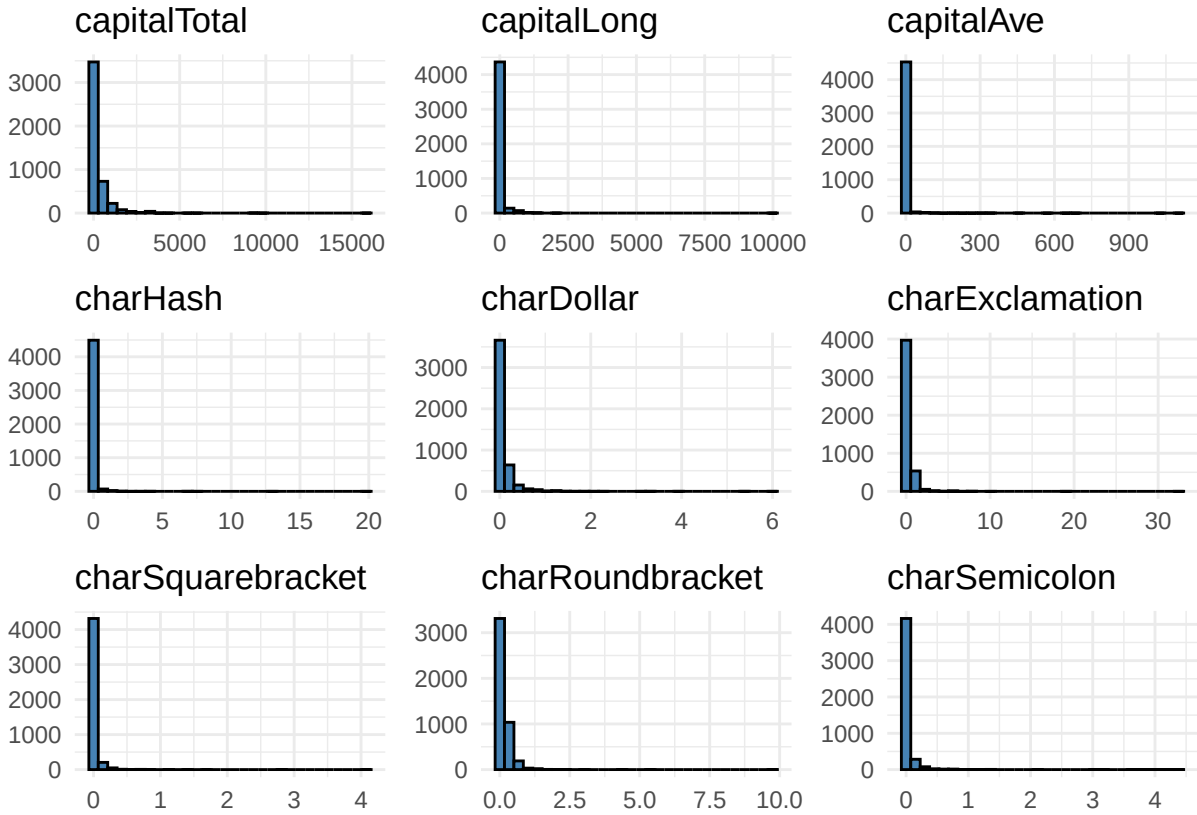
2. This dataset has 4601 observations and 57 features. The features are all numeric while the response is categorical, making this a classification problem. A table of some of the numeric features are presented below indicating a heterogeneous scale for the numeric features.

Table 5: Numeric features with scale summary

| Feature | Min | Max   |
|---------|-----|-------|
| num3d   | 0   | 42.81 |
| meeting | 0   | 14.28 |
| telnet  | 0   | 12.50 |
| labs    | 0   | 5.88  |
| lab     | 0   | 14.28 |

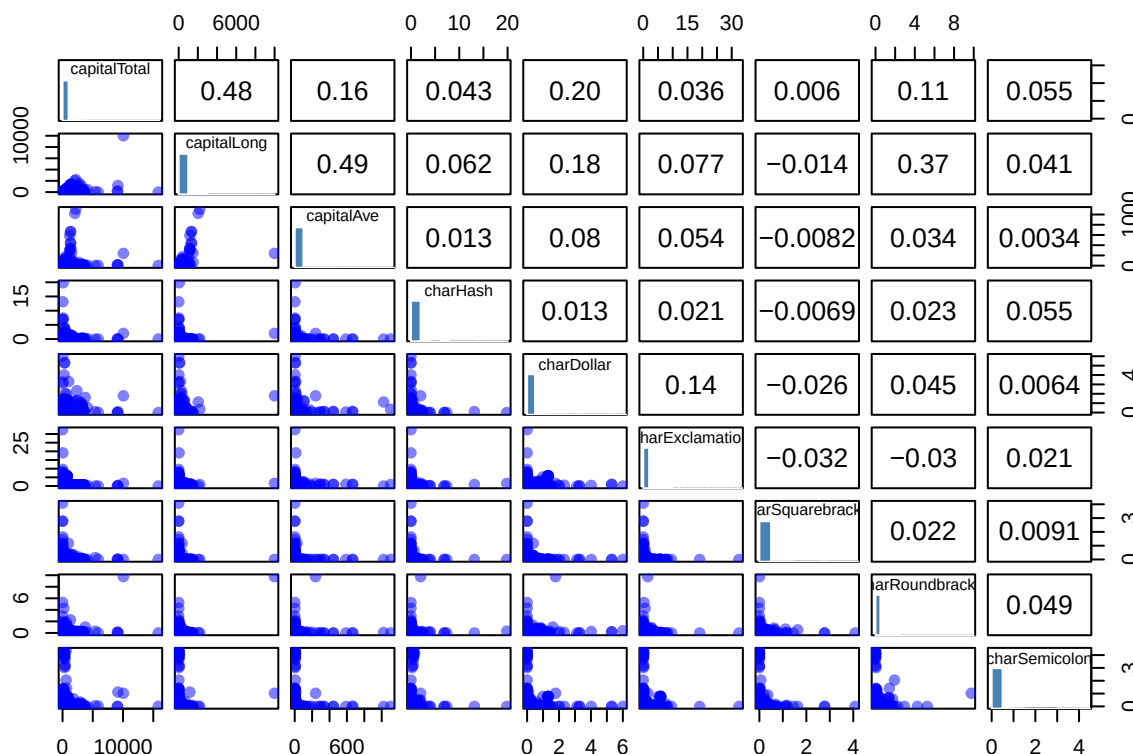
3. The index  $\kappa$  for this dataset is 80.719 this indicates that the number of samples are more than the observed features (an overdetermined system). This suggests that machine learning estimators are likely to be stable, and the risk of overfitting is relatively low. The large  $\kappa$  value implies there is sufficient information in the data to reliably estimate model parameters.

4. These histograms show that some features are extremely right-skewed, with most emails having values clustered very close to zero. A small number of emails contain unusually high counts of characters or capital letters, creating long tails in several distributions. Overall, this suggests that while most messages use these features, a few outliers may strongly influence how a classifier identifies spam-like behaviour.



5. The pairwise correlations among the nine features are all quite small, with the strongest relationship around 0.48 between capitalTotal and capitalLong. This indicates some mild association but nothing close to the levels (e.g.,  $>0.8$ ) that raise multicollinearity concerns. The scatterplots confirm this: the relationships are weak, scattered, and mostly nonlinear, suggesting that each feature contributes different information about character usage in emails. Overall, the low correlations mean collinearity is minimal, and multicollinearity is not

an issue. These predictors can coexist comfortably in a model without causing instability or inflated variance.



## Leukemia Data

- Below is a table of a few of the rows in the dataset.

| x.2462     | x.2510    | x.2226     | x.525      | x.194     |
|------------|-----------|------------|------------|-----------|
| -1.4140952 | 0.9626709 | 0.0920567  | -0.4909099 | 0.3759408 |
| -0.8169876 | 0.7307726 | 0.7381219  | -0.6230732 | 1.2540752 |
| -0.3412622 | 0.8111270 | -0.0531827 | 0.3076012  | 0.3591718 |

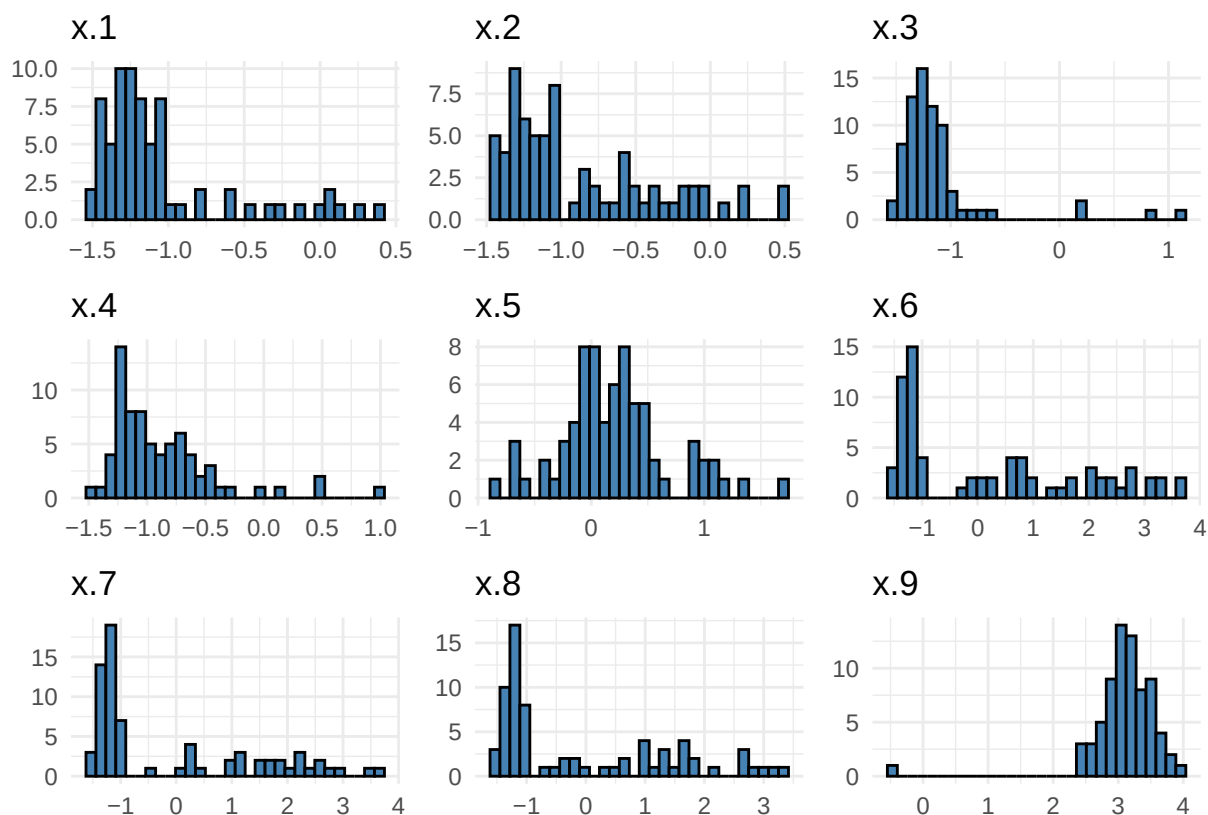
- This dataset has 72 observations and 3571 features. There is only 1 different data type (numeric) among the features. Below is a table of some of the numeric features with their scales, indicating a heterogeneous scale.

Table 7: Numeric features with scale summary

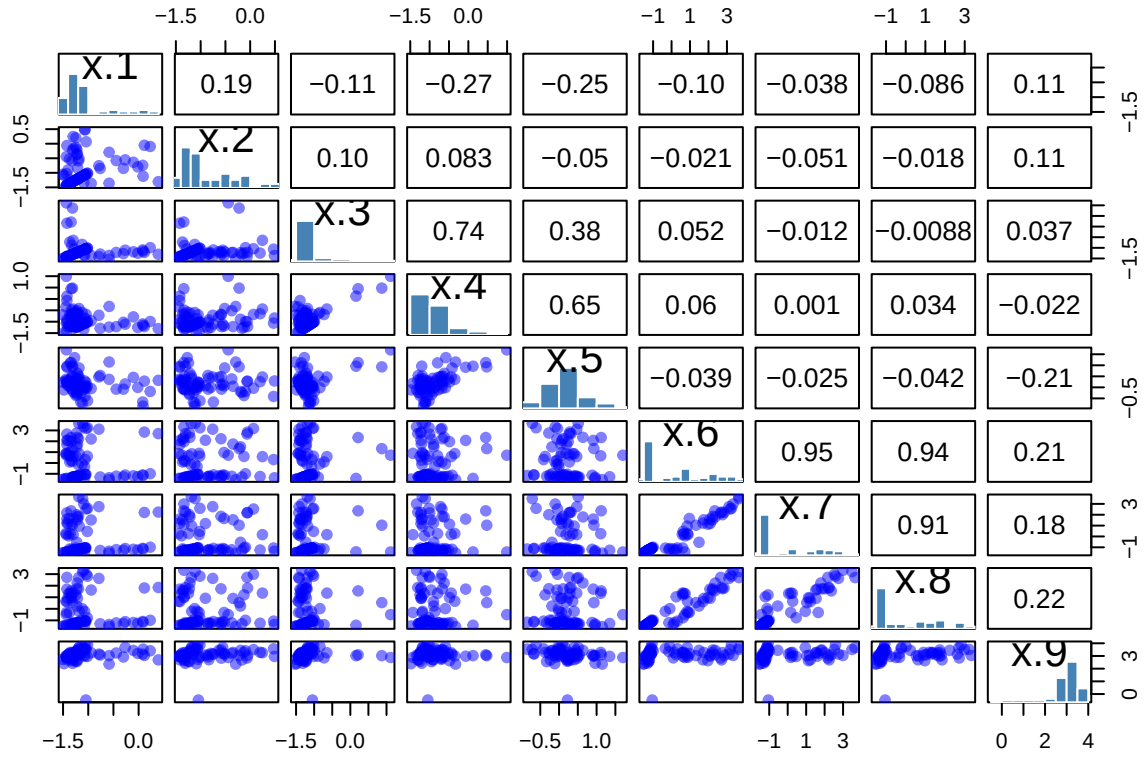
| Feature | Min        | Max      |
|---------|------------|----------|
| x.2985  | -1.4243923 | 1.342014 |
| x.1841  | -1.2545862 | 1.350872 |
| x.1141  | -0.2267222 | 2.225467 |
| x.3370  | -1.4816277 | 2.793430 |
| x.1252  | -1.2463837 | 0.774674 |

3. The index  $\kappa$  for this dataset is 0.02 this indicates that the number of samples are way less than the observed features (an underdetermined system). This suggests that machine learning estimators are likely to be unstable and prone to overfitting, unless strong regularisation or dimensionality reduction is applied. The small  $\kappa$  value implies that the data contain limited information for reliably estimating model parameters.

4. These histograms show that most of the variables are right-skewed, with many observations clustered on the left and a long tail extending to the right. A few variables (like x.5 and x.9) appear more balanced or centered, while others have clear outliers stretching the distribution. Overall, the features vary in shape, but many display skewness that may benefit from transformation depending on the modelling approach.



5. The correlations here are mostly very small, which means the variables are only weakly related to one another. A few pairs such as X.3–X.4 and X.6–X.7 show moderate positive relationships, but nothing approaching the high correlations (e.g.,  $>0.8$ ) that typically cause multicollinearity issues. Overall, the scatterplots look diffuse rather than tightly linear, so collinearity is minimal and multicollinearity is not a practical concern for modelling with these predictors.



## Prostate Data

1. Below is a table of a few of the rows in the dataset.

|      | X214065_s_at | X217522_at | X221420_at | X215861_at | X202735_at |
|------|--------------|------------|------------|------------|------------|
| PG13 | -0.3241601   | -0.3375143 | -0.3563790 | -0.3141736 | -0.1278737 |
| PG15 | -0.2696562   | -0.2841149 | -0.2764088 | -0.2298867 | -0.1214101 |
| PG37 | -0.2503474   | -0.2518213 | -0.2638858 | -0.2083398 | -0.1121377 |

2. This dataset has 79 observations and 500 features. All features are numeric, (type homogeneity). A table with the scales of some of these numeric features are given below. These scales indicate scale heterogeneity.

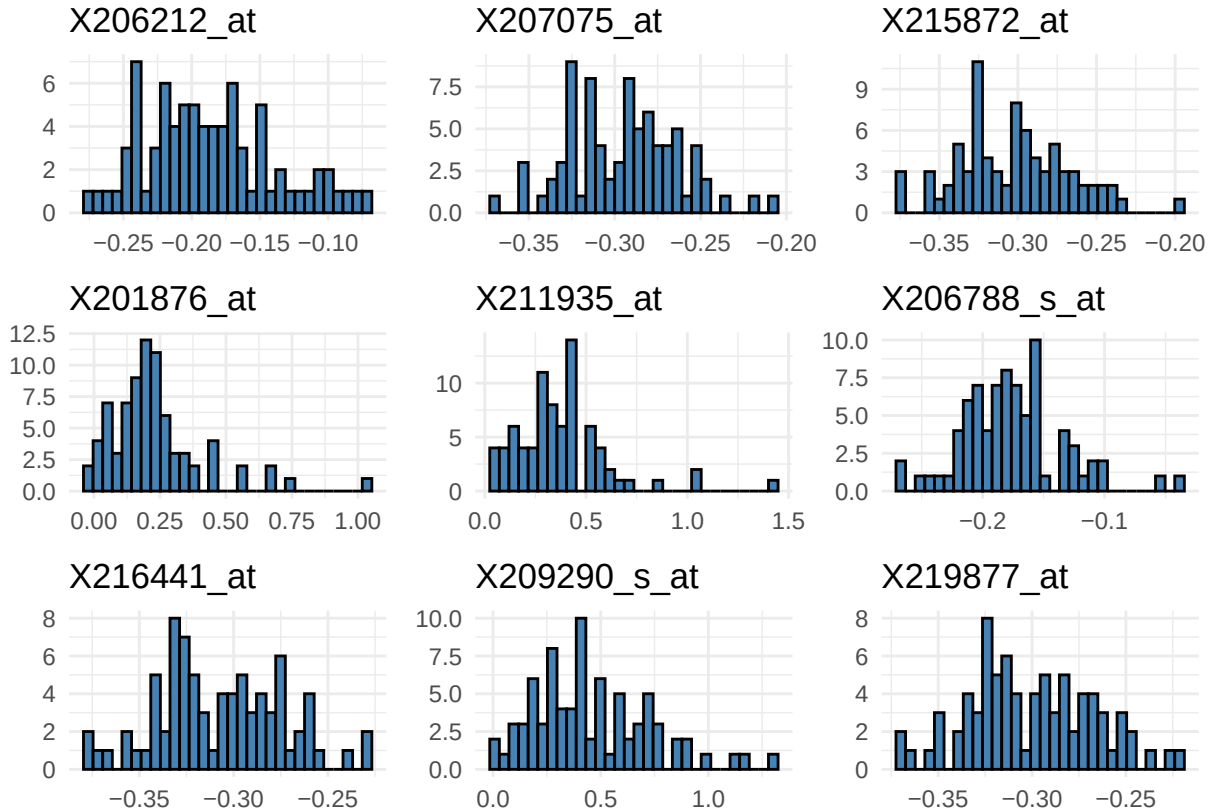
Table 9: Numeric features with scale summary

| Feature      | Min        | Max        |
|--------------|------------|------------|
| X206465_at   | -0.3627913 | -0.2106258 |
| X213166_x_at | 0.0439634  | 1.6443740  |
| X215738_at   | -0.3446781 | -0.1440000 |
| X213447_at   | -0.1189823 | 0.4761519  |
| X220488_s_at | -0.2557114 | -0.0680252 |

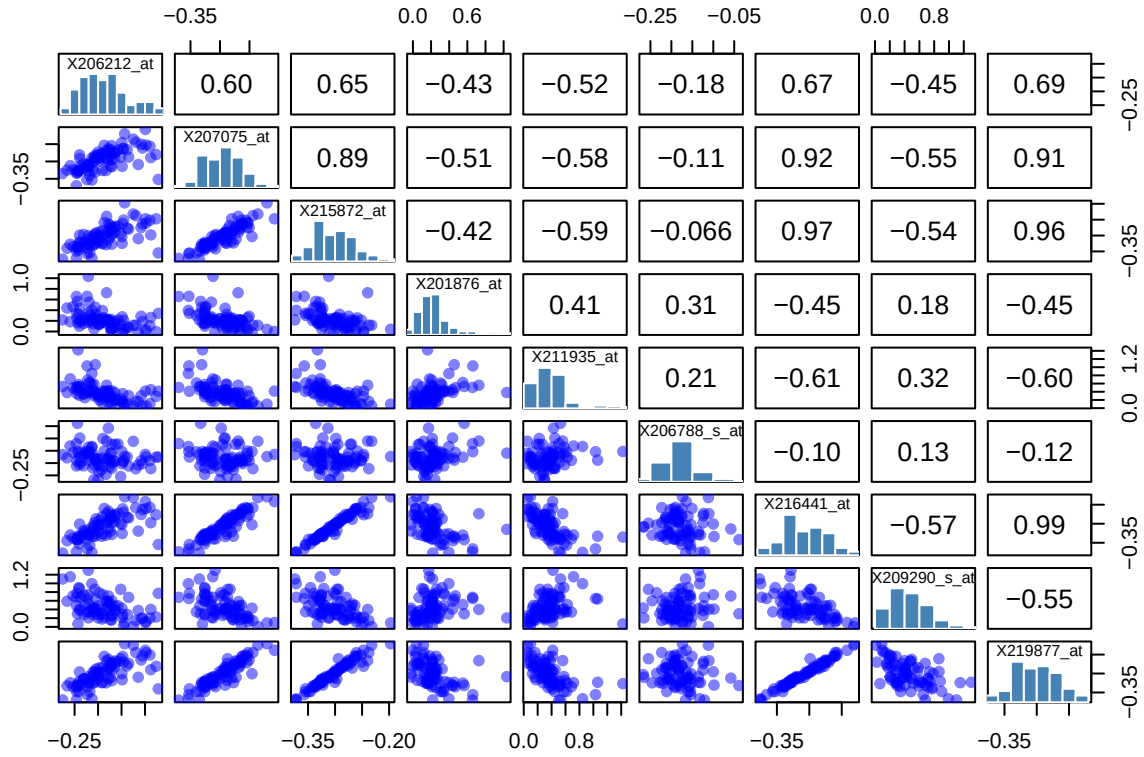
3. The index  $\kappa$  for this dataset is 0.158 this indicates that the number of samples are less than the observed features (an underdetermined system). This suggests that machine learning estimators are likely to be

unstable and prone to overfitting, unless strong regularisation or dimensionality reduction is applied. The small  $\kappa$  value implies that the data contain limited information for reliably estimating model parameters.

4. These histograms show that some gene expression features have their own distinct distribution, with most values clustering within a narrow range and only a few observations stretching into the tails. Some genes appear more symmetrically distributed, while others display noticeable right-skewness, suggesting occasional higher expression levels. Overall, the plots highlight that the genes vary in both spread and shape, which may help explain differences in how strongly each one contributes to downstream modelling.



5. These pairwise plots show that several gene-expression features are strongly correlated, with many correlation values above 0.9. The scatterplots for these pairs form very tight linear patterns, confirming that the variables move almost identically. Such high correlations indicate serious collinearity, meaning these genes carry nearly the same information. When used together in a regression or similar model, they would create multicollinearity, leading to unstable coefficient estimates and difficulty separating each gene's individual effect.



## Colon Data

1. Below is a table of a few of the rows in the dataset.

| X414      | X462       | X178     | X525      | X194      |
|-----------|------------|----------|-----------|-----------|
| 0.9183690 | 0.6105782  | 1.048785 | 0.9845771 | 0.9797156 |
| 0.2139728 | -0.6466094 | 1.524639 | 0.9453600 | 0.6979945 |
| 1.0127060 | 0.3355917  | 0.593305 | 1.2700206 | 1.6433465 |

2. This dataset has 62 observations and 2000 features. The features are all numeric. The table below shows some of the features and their scales, indicating scale heterogeneity.

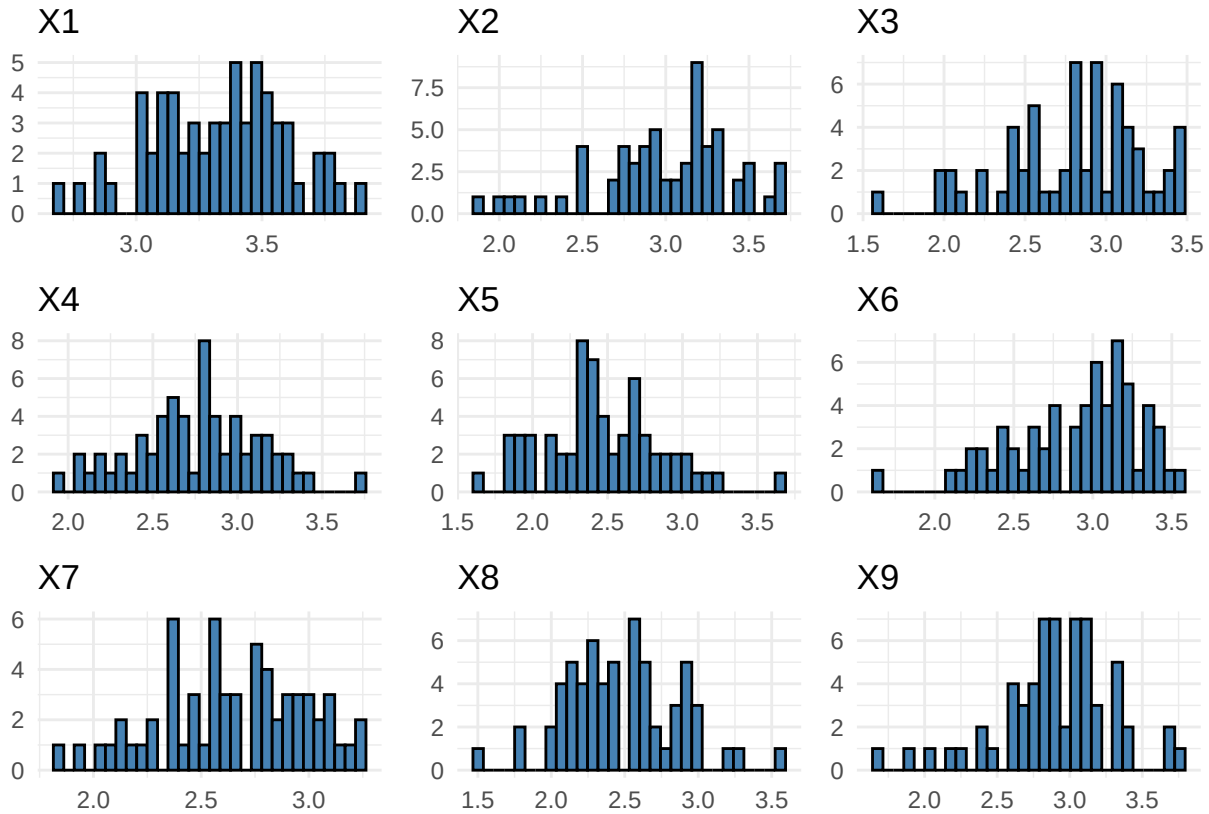
Table 11: Numeric features with scale summary

| Feature | Min       | Max        |
|---------|-----------|------------|
| X937    | -1.421323 | 0.6851193  |
| X1841   | -3.256031 | 0.1303733  |
| X1141   | -1.511940 | 0.9222931  |
| X1322   | -1.887731 | -0.0958678 |
| X1252   | -1.620780 | 0.3181653  |

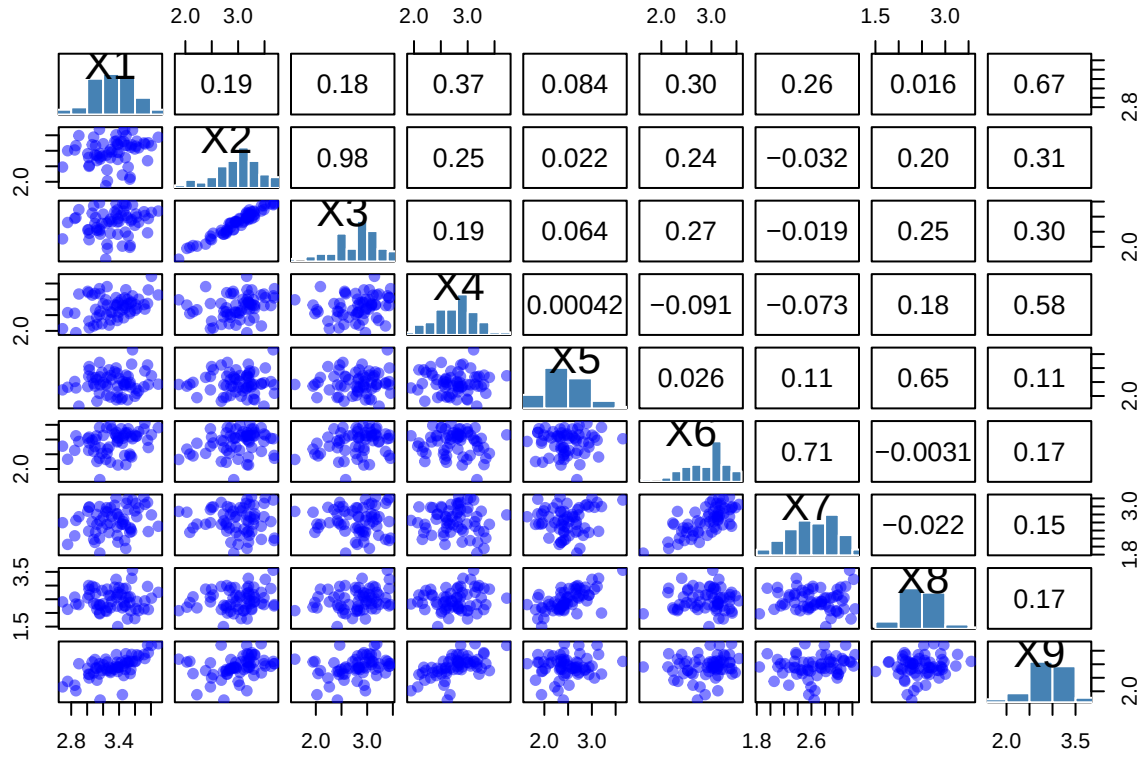
3. The index  $\kappa$  for this dataset is 0.031 this indicates that the number of samples are less than the observed features (an underdetermined system). This suggests that machine learning estimators are likely to be

unstable and prone to overfitting, unless strong regularisation or dimensionality reduction is applied. The small  $\kappa$  value implies that the data contain limited information for reliably estimating model parameters.

4. These histograms show that all nine variables follow roughly bell-shaped distributions, though each with slightly different centers and spreads. Some variables, like X5 and X7, appear more tightly concentrated, while others such as X2 and X6 are more spread out across their ranges. Overall, the features look reasonably normal, with no extreme skewness or heavy outliers, suggesting they're suitable for modelling without major transformation.



5. This plot shows that most variable pairs have very low correlations, meaning the predictors are largely independent of one another. Only a couple of relationships stand out, like X1 to X2 and X6 to X7 but these correlations are moderate rather than extreme. The relation between X2 and X3 stand out because of the nature of the scatter plot as well as the high correlation coefficient, indicating some multicollinearity. Overall, most variables do not exhibit the strong linear relationships that cause multicollinearity, even though some of them do. These variables that do exhibit a high multicollinearity could destabilize the model.



#### 4. Investigating the Bayes Risk $R^*$ in Regression

Let  $X$  and  $Y$  be two continuous random variables with joint probability density function

$$p(x, y) = \frac{1}{2\pi\sqrt{2\pi\frac{9}{\pi^2}}} \exp \left\{ -\frac{\pi^2}{18} \left[ y - \frac{\pi}{2}x - \frac{3\pi}{4} \cos\left(\frac{\pi}{2}(1+x)\right) \right]^2 \right\}.$$

It can be shown that the conditional density of  $Y$  given  $X$  is given by

$$p_1(y | x) = \frac{1}{\sqrt{2\pi\frac{9}{\pi^2}}} \exp \left\{ -\frac{\pi^2}{18} \left[ y - \frac{\pi}{2}x - \frac{3\pi}{4} \cos\left(\frac{\pi}{2}(1+x)\right) \right]^2 \right\}, \quad -\infty < x < \infty,$$

while the marginal density of  $X$  is given by

$$p_2(x) = \frac{1}{2\pi}, \quad 0 \leq x < 2\pi.$$

##### 1. Find and write down the expression for $\mathbb{E}[Y | X]$

Notice that the conditional distribution of  $Y$  given  $X$  can be rewritten to get:

$$\frac{1}{\sqrt{2\pi\frac{9}{\pi^2}}} \exp \left\{ -\frac{\pi^2}{18} \left[ y - \frac{\pi}{2}x - \frac{3\pi}{4} \cos\left(\frac{\pi}{2}(1+x)\right) \right]^2 \right\} = \frac{1}{\frac{3}{\pi}\sqrt{2\pi}} \exp \left\{ -\frac{\left( y - \left( \frac{\pi x}{2} + \frac{3\pi \cos \frac{\pi(1+x)}{2}}{4} \right) \right)^2}{2\left(\frac{9}{\pi^2}\right)} \right\}$$

which is exactly the normal distribution with mean  $\left(\frac{\pi x}{2} + \frac{3\pi \cos \frac{\pi(1+x)}{2}}{4}\right)$  and variance  $\frac{9}{\pi^2}$

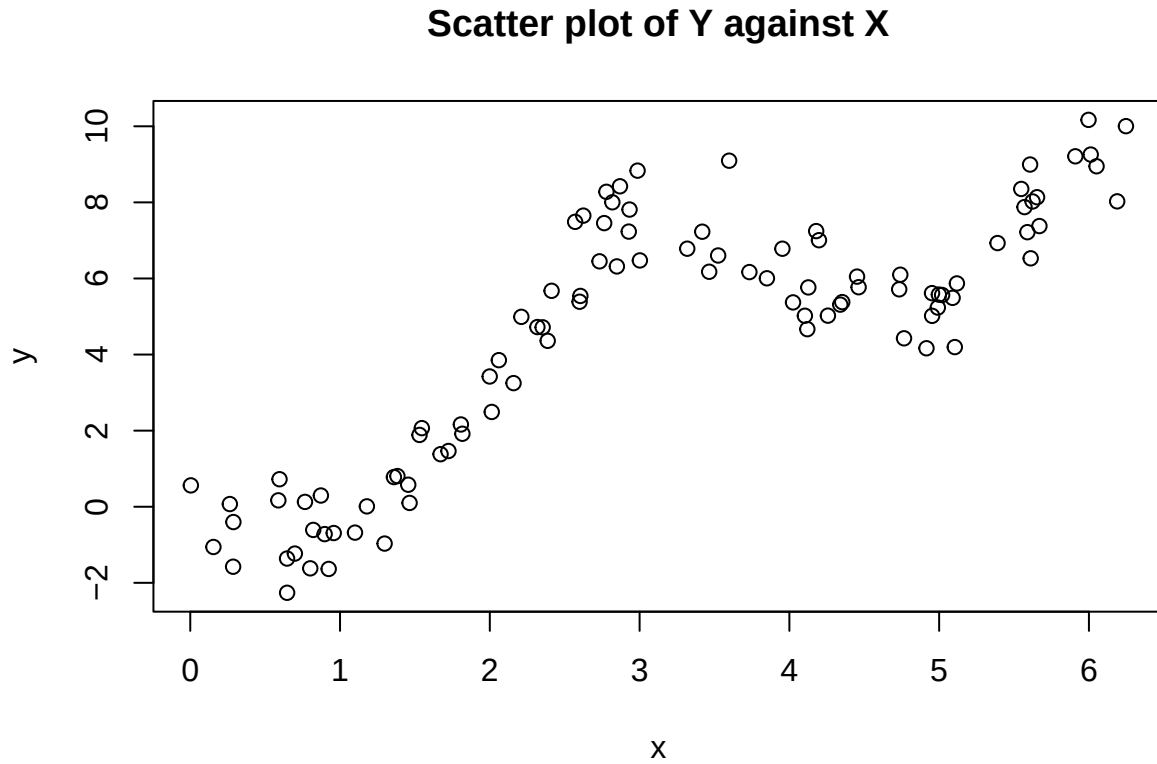
## 2. Generate an iid sample

Below is a table of the first 10  $(x, y)$  pairs that were randomly generated.

Table 12: First 10  $(x, y)$  pairs

|    | x         | y         |
|----|-----------|-----------|
| 1  | 1.8069028 | 2.162056  |
| 2  | 4.9530672 | 5.017812  |
| 3  | 2.5696778 | 7.488250  |
| 4  | 5.5481620 | 8.352669  |
| 5  | 5.9091302 | 9.210446  |
| 6  | 0.2862399 | -1.576140 |
| 7  | 3.3181846 | 6.782776  |
| 8  | 5.6072342 | 8.994681  |
| 9  | 3.4647684 | 6.176293  |
| 10 | 2.8689950 | 8.422993  |

## 3. Draw the scatter plot of the iid sample



The scatter plot shows a clear curved pattern, where  $Y$  increases and decreases smoothly as  $X$  varies. This aligns with the theoretical conditional mean, a sinusoidal function rather than a straight line. Although the

individual points are scattered because of noise, the overall trend is visibly nonlinear, suggesting that a simple linear regression would not capture the true relationship between  $X$  and  $Y$ .

**Question 4: Regression Analysis and Bayes Risk**